

Unlocking the Potential of Text-to-Image Diffusion with PAC-Bayesian Theory

Eric Hanchen Jiang, Yasi Zhang, Zhi Zhang, Yixin Wan, Andrew Lizarraga,
Shufan Li, and Ying Nian Wu
University of California, Los Angeles
ericjiang0318@g.ucla.edu

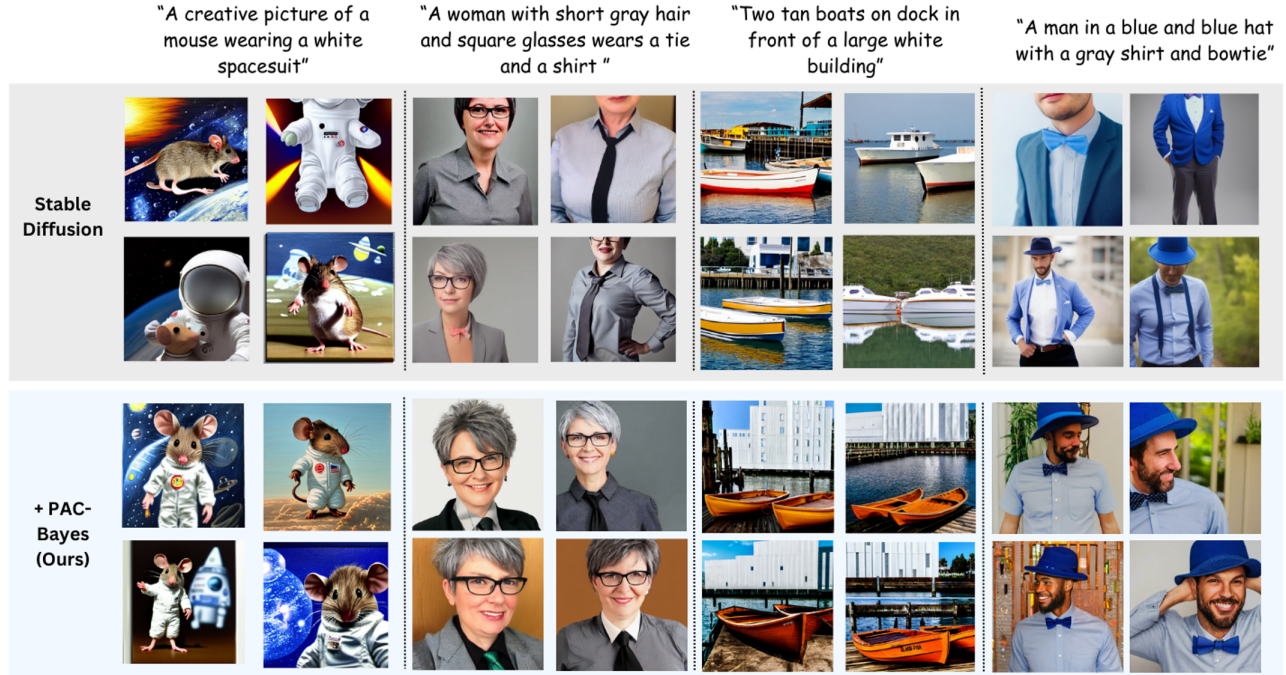


Figure 1. Diffusion models often struggle to accurately represent multiple objects in the input text. We identify the root causes of these challenges and introduce a training-free solution to address them by using PAC-Bayesian Theory. Here, we show some qualitative image results based on our model compared to the original Stable Diffusion model. For instance, when prompted to depict “a mouse wearing a white spacesuit”, Stable Diffusion fails to separately associate individual modifiers with each component, therefore neglecting some of the descriptors (i.e. only generating a mouse or a spacesuit).

Abstract

Text-to-image (T2I) diffusion models have revolutionized generative modeling by producing high-fidelity, diverse, and visually realistic images from textual prompts. Despite these advances, existing models struggle with complex prompts involving multiple objects and attributes, often misaligning modifiers with their corresponding nouns or neglecting certain elements. Recent attention-based methods have improved object inclusion and linguistic binding, but still face challenges such as attribute misbinding and a lack of robust generalization guarantees. Leveraging the

PAC-Bayes framework, we propose a Bayesian approach that designs custom priors over attention distributions to enforce desirable properties, including divergence between objects, alignment between modifiers and their corresponding nouns, minimal attention to irrelevant tokens, and regularization for better generalization. Our approach treats the attention mechanism as an interpretable component, enabling fine-grained control and improved attribute-object alignment. We demonstrate the effectiveness of our method on standard benchmarks, achieving state-of-the-art results across multiple metrics. By integrating custom priors into the denoising process, our method enhances image quality and addresses long-standing challenges in T2I diffusion